

Machine Learning for Return Prediction

BUSI 722: Data-Driven Finance II

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Standardizing Features

Why Standardize?

- Raw features have very different scales: marketcap is in thousands, roe is a fraction, momentum can be $\pm 100\%$.
- Many models (linear regression, neural networks) are sensitive to scale.
- Standardizing puts all features on a common scale and tames outliers.

Cross-Sectional Standardization

Standardize each feature **within each month**, not over the full panel.

For feature x in month t :

$$z_{i,t} = \frac{x_{i,t} - \bar{x}_t}{\sigma_{x,t}}$$

- Each month's cross section has mean 0 and standard deviation 1.
- Removes time-series level shifts (e.g., all P/E ratios rising over time).
- Ensures the model learns **which stocks are cheap relative to their peers this month**, not which months had low valuations overall.

Ranks as an Alternative

Instead of z-scores, rank stocks 1 through n each month, then scale to $[0, 1]$:

$$\text{rank}_{i,t} = \frac{\text{rank of } x_{i,t} \text{ among all stocks in month } t}{n_t}$$

- Completely eliminates outliers — every stock gets a value in $[0, 1]$.
- Preserves only the **ordering** of stocks, not the magnitudes.
- Robust to skewed distributions; can also compute z-scores of ranks for a hybrid approach.

Standardizing the Target

Predicting Relative Returns

We care more about **which stocks will outperform** than about predicting exact return magnitudes.

Stocks move up and down together to a large extent, and the common (market) movement is hard to predict. We don't need to predict it if we are just predicting which stocks will do relatively well and which will do relatively poorly.

This suggests we should think carefully about what target variable we give the model.

Ranks as the Target

Replace the raw return $r_{i,t}$ with its **cross-sectional rank** each month:

$$y_{i,t} = \frac{\text{rank of } r_{i,t} \text{ among all stocks in month } t}{n_t}$$

- The model learns to predict **relative performance**, not absolute returns.
- Removes the influence of market-wide shocks and reduces the impact of return outliers.
- Scaled to $[0, 1]$: top stock ≈ 1 , bottom stock ≈ 0 .

Comparison of Targets

	Raw Returns	Ranked Returns
Scale	varies by month	always $[0, 1]$
Outliers	can be extreme	bounded
Market effect	included	removed
Interpretation	absolute return	relative ranking
Loss function	MSE on returns	MSE on ranks

Using ranked returns is common in quantitative equity research. It focuses the model on what actually drives portfolio construction: **selecting the right stocks relative to their peers.**

Models: A Review

The Prediction Problem

We have a panel of stock-months. For each stock i in month t :

- Features: $\mathbf{x}_{i,t-1}$ (known at the start of month t)
- Target: $y_{i,t}$ (return or return rank during month t)

We want to learn a function f such that

$$y_{i,t} \approx f(\mathbf{x}_{i,t-1})$$

- Train on past data, predict on future data (no look-ahead bias).
- The function f can be linear, a tree ensemble, a neural network, etc.

$$f(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$$

- Simple, interpretable, fast to estimate.
- Each coefficient β_j measures the marginal effect of feature j .
- Assumes the effect of each feature is **additive and linear** — no interactions, no diminishing effects.

Regularization

Add a penalty to prevent overfitting: $\min \sum (y_i - f(\mathbf{x}_i))^2 + \text{penalty}$

- **Ridge** (L_2): $\text{penalty} = \lambda \sum \beta_j^2$. Shrinks coefficients toward zero, keeps all features.
- **Lasso** (L_1): $\text{penalty} = \lambda \sum |\beta_j|$. Drives some coefficients to exactly zero (feature selection).
- **Elastic net**: combines L_1 and L_2 penalties. Balances shrinkage and sparsity.

The hyperparameter λ controls the strength of regularization. Larger λ = simpler model.

Limitations of Linearity

- A linear model says: “if momentum goes up by 1, expected return goes up by β .”
- But the real relationship may be **nonlinear**:
 - Momentum may help only above a threshold.
 - Value may matter more for small stocks than large stocks (interaction).
 - The effect of leverage may reverse at high levels.
- We need models that can capture these patterns automatically.

Decision Trees

A decision tree partitions the feature space into rectangular regions using a sequence of binary splits.

- At each node, pick the feature and threshold that best reduces prediction error.
- Naturally captures **nonlinearities** and **interactions**.
- A single tree is interpretable but tends to overfit.

Solution: combine many trees into an **ensemble**.

Random Forest

- Grow B trees, each on a **bootstrap sample** of the training data.
- At each split, consider only a **random subset** of features.
- Prediction = average of all B trees.

Key properties:

- Reduces variance through averaging (bagging).
- Random feature selection decorrelates trees.
- Robust to outliers, requires little tuning, and hard to overfit by adding more trees.

Gradient Boosting (GBM)

Instead of averaging independent trees, build trees **sequentially**, each correcting the errors of the previous ones.

1. Start with a constant prediction (e.g., the mean).
2. Fit a shallow tree to the **residuals** (prediction errors).
3. Add the new tree's prediction (scaled by learning rate η) to the running total; repeat for B rounds.

Key hyperparameters: number of trees B , learning rate η , tree depth, minimum samples per leaf.

LightGBM

- Microsoft's implementation of gradient boosting, optimized for speed and scale.
- Uses **histogram-based** splitting and **leaf-wise** growth for faster, more accurate tree building.
- Widely used in quantitative finance and Kaggle competitions.

In practice, LightGBM often outperforms random forests on tabular financial data.

Histogram-Based Splitting

Standard gradient boosting evaluates every unique value of every feature to find the best split — slow for large datasets.

LightGBM's approach:

- Bin each continuous feature into K discrete buckets (e.g., $K = 255$) once before training begins.
- At each node, scan only K bin boundaries instead of thousands of unique values.
- Speeds up split-finding dramatically with negligible loss in accuracy.

The histogram also enables **efficient memory use**: store integer bin indices rather than raw floats, and accumulate gradient sums per bin instead of per observation.

Leaf-Wise Growth

Most gradient boosting implementations (e.g., scikit-learn) grow trees **level-wise**: split every node at a given depth before going deeper.

LightGBM grows leaf-wise instead:

- At each step, find the single leaf among *all current leaves* that yields the largest reduction in loss, and split only that leaf.
- This produces **asymmetric trees** that concentrate depth where the data is most complex.
- For the same number of splits, leaf-wise growth typically achieves lower training loss than level-wise growth.

Trade-off: leaf-wise trees can overfit on small datasets; the *num_leaves* hyperparameter (rather than depth) is the primary complexity control.

Random Fourier Features

A way to add nonlinearity to a linear model without the complexity of trees or neural networks.

Idea: transform the original k features into a much larger set of D random nonlinear features, then fit a linear model on the transformed features.

$$\phi(\mathbf{x}) = \sqrt{\frac{2}{D}} \begin{pmatrix} \cos(\mathbf{w}'_1 \mathbf{x} + b_1) \\ \vdots \\ \cos(\mathbf{w}'_D \mathbf{x} + b_D) \end{pmatrix}$$

where $\mathbf{w}_j \sim N(0, \gamma^2 I)$ and $b_j \sim \text{Uniform}(0, 2\pi)$ are drawn **randomly and fixed**.

Kernels and the RBF Kernel

A **kernel** $k(\mathbf{x}, \mathbf{z})$ is a similarity measure between two feature vectors.

Kernel regression predicts by taking a weighted average of training targets, where the weights are similarities to $\mathbf{x}$:

$$f(\mathbf{x}) = \sum_{i=1}^n \alpha_i k(\mathbf{x}, \mathbf{x}_i)$$

The **RBF (radial basis function) kernel** — also called the Gaussian kernel — is:

$$k(\mathbf{x}, \mathbf{z}) = \exp(-\gamma^2 \|\mathbf{x} - \mathbf{z}\|^2)$$

- Near 1 when $\mathbf{x}$ and $\mathbf{z}$ are close; near 0 when they are far apart.
- γ controls how quickly similarity decays — large γ means only very nearby points influence the prediction.

RFF as an RBF Kernel Approximation

A key mathematical result (Bochner's theorem) implies that for the RBF kernel with $\mathbf{w} \sim N(0, \gamma^2 I)$:

$$k(\mathbf{x}, \mathbf{z}) = E_{\mathbf{w}} [\cos(\mathbf{w}'(\mathbf{x} - \mathbf{z}))]$$

Expanding using the cosine difference formula:

$$k(\mathbf{x}, \mathbf{z}) = E_{\mathbf{w}} [\cos(\mathbf{w}'\mathbf{x}) \cos(\mathbf{w}'\mathbf{z}) + \sin(\mathbf{w}'\mathbf{x}) \sin(\mathbf{w}'\mathbf{z})]$$

Approximate with $D/2$ random draws $\mathbf{w}_1, \dots, \mathbf{w}_{D/2}$:

$$k(\mathbf{x}, \mathbf{z}) \approx \phi(\mathbf{x})' \phi(\mathbf{z}), \quad \phi(\mathbf{x}) = \frac{1}{\sqrt{D/2}} \begin{pmatrix} \cos(\mathbf{w}'_1 \mathbf{x}) \\ \sin(\mathbf{w}'_1 \mathbf{x}) \\ \vdots \\ \cos(\mathbf{w}'_{D/2} \mathbf{x}) \\ \sin(\mathbf{w}'_{D/2} \mathbf{x}) \end{pmatrix}$$

Phase-Shift Equivalence

The formulation on the earlier slide used only cosines with a random phase shift $b_j \sim \text{Uniform}(0, 2\pi)$. This is equivalent, via the identity:

$$\cos(\mathbf{w}'\mathbf{x} + b) \cos(\mathbf{w}'\mathbf{z} + b) = \frac{1}{2} \cos(\mathbf{w}'(\mathbf{x} - \mathbf{z})) + \frac{1}{2} \cos(\mathbf{w}'(\mathbf{x} + \mathbf{z}) + 2b)$$

Averaging over $b \sim \text{Uniform}(0, 2\pi)$, the second term vanishes, leaving:

$$E_b [\cos(\mathbf{w}'\mathbf{x} + b) \cos(\mathbf{w}'\mathbf{z} + b)] = \frac{1}{2} \cos(\mathbf{w}'(\mathbf{x} - \mathbf{z}))$$

This is *half* of what sine+cosine pairs give, hence the $\sqrt{2}$ in the $\sqrt{2/D}$ prefactor.

Intuition: $\cos(\mathbf{w}'\mathbf{x} + b)$ with random b is a random mixture of $\cos(\mathbf{w}'\mathbf{x})$ and $\sin(\mathbf{w}'\mathbf{x})$ — one random-phase cosine does the work of one sine-cosine pair, up to the factor of 2.

Why Random Fourier Features Work

- The cosine transformation creates **nonlinear combinations** of the original features, capturing interactions and curved relationships.
- The random weights $\mathbf{w}_j$ are not learned — only the final linear coefficients are estimated.
- The bandwidth γ controls how “wiggly” the fitted function is: large γ allows rapid local variation, small γ produces smoother global fits.

Advantages: fast to fit (just linear regression on transformed features), no neural network training, easy to regularize with ridge or lasso.

Neural Networks

A neural network is a sequence of linear transformations interleaved with nonlinear **activation functions**.

For a network with one hidden layer of h units:

$$f(\mathbf{x}) = \mathbf{w}'_2 \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + b_2$$

- $\mathbf{W}_1$ is an $h \times k$ weight matrix, $\sigma(\cdot)$ is a nonlinear activation (ReLU, tanh, etc.).
- The hidden layer learns **nonlinear features** from the raw inputs.
- Adding more layers = deeper network = more complex representations.

Training Neural Networks

- Minimize a loss function (e.g., MSE) using **stochastic gradient descent** (SGD) or variants like Adam.
- Process data in **mini-batches**; one pass through the full training set = one **epoch**.
- Training continues for many epochs; monitor validation loss to detect overfitting.

Regularization:

- **Dropout**: randomly zero out a fraction of hidden units during training.
- **Early stopping**: stop training when validation loss stops improving.
- **Weight decay** (L_2 penalty on weights).

Neural Networks vs. Other Models

	Linear	Trees	RFF	Neural Net
Nonlinearity	no	yes	yes	yes
Interactions	no	yes	yes	yes
Interpretable	yes	moderate	no	no
Tuning effort	low	moderate	low	high
Training speed	fast	fast	fast	slow

No single model dominates. In practice, **ensembles** that combine predictions from multiple model types often perform best.